Domain Shift-Aware Training-Free Adaptation for Open-Vocabulary Segmentation in Robotic Perception

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***Abstract*— Robust and reliable perception is essential for autonomous robots operating in real-world environments, particularly in long-term missions where environmental conditions may change significantly over time. Although recent advances in Visual Foundation Models (VFMs) have improved open-vocabulary semantic segmentation, these models can still suffer from domain shift, which can significantly degrade performance if they are not adapted to the current environment. In this paper, we propose a training-free continual test-time adaptation (TF-CTTA) approach for open-vocabulary segmentation that leverages temporal coherence across consecutive frames via a domain shift detector that triggers adaptation when needed. We validate our approach on a benchmark including indoor and outdoor environments, demonstrating that our method improves segmentation performance under domain shift while remaining more efficient, making the training-free adaptation more practical and feasible for long-term, real-world deployments.**

I. Introduction

For autonomous robots operating in real-world environments, robust perception is essential, especially during long-term missions, such as last-mile delivery, service robotics, or mobile assistance [1]. In these scenarios, a robot must navigate and understand scenes reliably across multiple and changing environments (e.g., alternating indoor and outdoor scenarios, new buildings, and varying illumination and weather). Such variability generates the well-known *domain shift* problem, which causes performance degradation if the model is not adapted to the current operational environment [2].

Recent advances in Visual Foundation Models (VFMs), particularly for open-vocabulary semantic segmentation [3], [4], have significantly improved robustness and generalization capabilities. Despite strong zero-shot capabilities, these models can still suffer notable performance degradation under domain shift. Changes in the scene type, appearance, or layout can lead to high segmentation quality in one context and noticeable degradation in another, making adaptation necessary during deployment [5].

Although domain adaptation for semantic segmentation has been widely explored, a core challenge still remains: how to adapt models online without introducing computational burden (e.g., operational cost or system complexity) while maintaining task performance. A potential solution is to rely on Continual test-time adaptation (CTTA) approaches [6], [7], consisting of fine-tuning the model online after each inference step. However, they can be impractical in real-world robotic deployments because online fine-tuning via backpropagation introduces computational overhead, latency, and memory demands on resource-constrained devices. A promising alternative is training-free adaptation, in which the model adjusts at inference time without gradient-based updates, relying instead on precomputed, lightweight mechanisms, such as adapters and model/configuration selection. In this direction, SemLA [8] proposes a library of low-rank adapters (LoRA) indexed with CLIP embeddings, where the most relevant adapters are dynamically retrieved and fused at test time based on their proximity to the target domain.

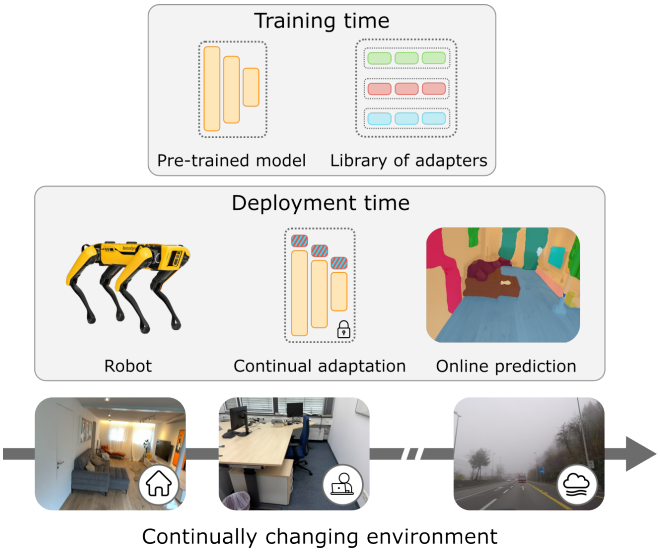


Fig. 1. The training-free continual test-time adaptation task considered in this paper.

While attractive, SemLA still faces practical limitations for real-world deployment. The method is applied on a frame-by-frame basis, requiring frequent adapters retrieval and fusion, which can increase memory usage, energy consumption, and runtime overhead. Furthermore, evaluations have mostly focused on general-purpose datasets and temporal consistency has been considered only outdoors, leaving indoor evaluation on sequential robot data completely unexplored.

To address these challenges, we propose a training-free continual test-time adaptation (TF-CTTA) approach for open-vocabulary segmentation in robotic perception. Our setting, illustrated in Fig. 1, applies a SemLA-style paradigm to a real-world mobile robot operating across indoor and outdoor

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environments. As [8], our TF-CTTA solution assumes a large library of precomputed adapters and exploits temporal coherence across consecutive frames via a domain shift detector that triggers adaptation when needed. We demonstrate that our solution makes the training-free adaptation more practical and feasible for long-term, real-world deployments.

In summary, our contributions are as follows:

- We analyze the performance of SemLA [8] on real-world robotic sequences, with a focus on indoor and alternating indoor-outdoor scenarios, to identify its practical limitations.
- We introduce an online domain shift detection method to guide the adaptation, with the purpose of reducing the amount of adapters to load in the device memory and mitigating the inference overhead caused by a per-frame adaptation, while maintaining semantic segmentation quality.
- We propose a continual test-time adaptation benchmark tailored to robotics that captures alternating and temporally correlated domain shifts, providing a comprehensive evaluation comparing our TF-CTTA approach with SemLA and other relevant baselines.

II. Related Works

This section reviews prior work most relevant to our approach, with an emphasis on continual test-time domain adaptation, methods for detecting and handling domain shift, and training-free adaptation.

Continual test-time adaptation (CTTA) is a challenging task that focuses on enabling models to adapt in real-time to dynamic, changing environments, without access to the source data. Recent methods focus on improving the stability and efficiency of adaptation. [9] introduces a memory-efficient architecture that consists of lightweight meta networks that adapt frozen original networks to the target domains by reducing the intermediate activation size used for backpropagation, and employs a self-distilled regularization method to prevent catastrophic forgetting and error accumulation. [10] proposes visual domain adapters that are added to a source-pretrained model to handle both domain-specific and domain-shared knowledge, aiming to simultaneously adapt to the current domain distribution while maintaining continual domain-shared knowledge over time. [11] proposes a distribution-aware tuning approach that selects and fine-tunes domain-specific and task-relevant parameters, together with a Parameter Accumulation Update (PAU) strategy that collects parameter updates to stabilize the selection. Moving toward more realistic deployment scenarios, [12] studies multi-modality test-time adaptation for semantic segmentation, leveraging multiple sensing modalities to improve robustness under domain shifts and highlighting the relevance of methods tailored for robotic perception.

Domain Shift Detection for CTTA is a key component in continual test-time adaptation for mitigating performance degradation due to domain shift between the source domain and target domain. Some methods detect domain shifts by leveraging the discrepancy between the teacher-student predictions. Hybrid-TTA [6] compares the consistency loss between teacher and student with a variable threshold, updated using an exponential moving average (EMA) of the loss at each time step. Similarly, [13] uses the consistency loss between teacher and student. In this case, the loss is computed from the predictions of the student model and a small decoder head attached to the first layers of the teacher. The resulting signal is then denoised through bin averaging and compared with a threshold to detect whether a shift has occurred. [7] models domain shift directions via Concept Activation Vectors (CAVs) and incorporates domain shift and class shift controllers to prevent categories from drifting towards other categories, preserving class-separability during adaptation.

Other approaches detect domain shifts by performing uncertainty estimation using a model trained on the source domain. [11] and [10] dynamically evaluate the impact of the domain shift by measuring the uncertainty of the model using Monte Carlo dropout. In ViDA [10], each sample is processed multiple times using dropout in the linear layers, and uncertainty is calculated as the variance across the outputs. Onda [14] instead, uses the confidence of a frozen source-trained model to detect shifts over time. To reduce noise, it adopts a sliding-window strategy and computes a weighted average within the window, detecting a shift when its value exceeds a predefined threshold. Similarly, [15] estimates uncertainty using the average class-wise confidence of a source model, which is compared with a threshold interval to identify domain changes. Likewise, [16] computes a threshold using an EMA by aggregating the maximum confidence values for each pixel obtained from augmented data.

These approaches are not applicable in our scenario, as deploying an additional segmentation model significantly compromises computational and memory efficiency. Instead, we propose a domain shift detection method based on CLIP features that, being already required within the training-free adaptation framework for adapters indexing, avoids redundant components, preserving efficiency.

Training-free domain adaptation methods adjust pre-trained models to new target domains through post-processing techniques, such as weight merging and parameter fusion, avoiding any additional training. Authors in [17] propose a model merging technique, including parameter merging and buffer merging for multi-target domain adaptation tasks, applicable to any single-target domain adaptation model. Most closely related to our work, [8] introduces SemLA, a training-free test-time framework intended for open-vocabulary segmentation that leverages a library of LoRA-based adapters indexed with Contrastive Language-Image Pretraining (CLIP) embeddings [18]. The CLIP embeddings guide adapter selection, and subsequently, the most relevant adapters are merged based on their proximity to the target domain in the embedding space.

Positioning our work. Our work sits between continual test-time adaptation (CTTA) and training-free adaptation (TF) for robotic perception. Similar to CTTA, it handles

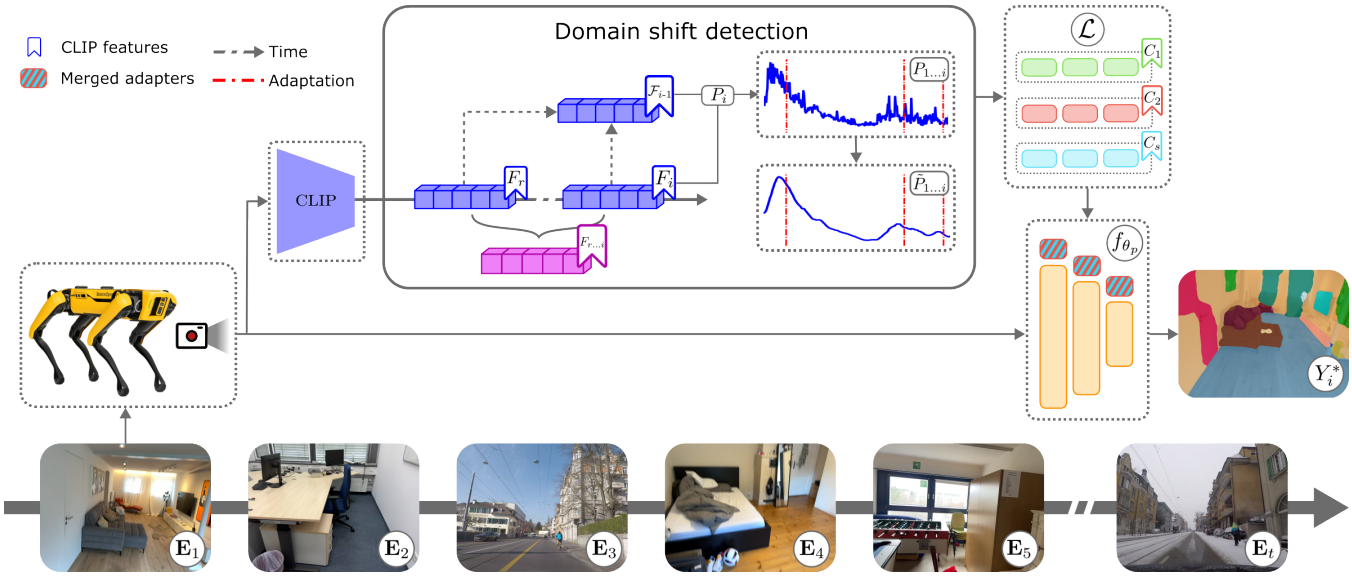


Fig. 2. General overview of the framework for training-free continual test-time adaptation, incorporating SemLA [8] and our domain shift detector.

sequential camera streams where domain shifts can evolve over time. However, our approach follows a training-free paradigm such as SemLA, in which adaptation is achieved by selecting from a library of pretrained adapters rather than performing gradient-based updates during deployment. By leveraging temporal coherence and incorporating a domain shift detector to avoid unnecessary adaptation, our approach is better suited to long-term robotic deployments.

III. Preliminaries

A. Problem formulation

In this paper, we consider the scenario of a robot performing open-vocabulary semantic segmentation from images. More formally, this task can be defined as $Y^* = f_{\theta_p}(I)$, where a neural network f correlates an image I to a dense mask Y^*, where each pixel is assigned to an object category. The network's parameters θ_p are pre-trained using a labeled *source* dataset $\mathcal{P} = (\mathbf{I}^p, \mathbf{Y}^p, \mathbf{O}^p)$ containing a list of images ($\mathbf{I}^p$) and their corresponding ground truth masks ($\mathbf{Y}^p$), where object categories come from $\mathbf{O}^p$. Inside the same mission, a robot needs to operate in a sequence of *target* environments $\mathcal{T} = [\mathbf{E}_t \mid t = 1, \dots, T]$ that can be both indoor or outdoor. Inside each environment $\mathbf{E}_t = \{\mathbf{I}_t, \mathbf{O}_t\}$, the robot is tasked to segment a sequence of RGB images $\mathbf{I}_t$ using a specific set of object categories $\mathbf{O}_t$. Note that we are working in open-vocabulary settings because the object classes used to initialize the model and those required at deployment time differ ($\mathbf{O}^p \neq \mathbf{O}_t, \forall\, t = 1, \dots, T$), as well as the classes required in different domains ($\exists\, y, j \in 1, ..., T | \mathbf{O}_y \neq \mathbf{O}_j$).

The task of training-free continual test-time adaptation (TF-CTTA) aims to adapt the neural network f_{θ_p}, initially pre-trained on the source dataset $\mathcal{P}$, to a sequence of target datasets $\mathcal{T}$, thus obtaining a new model f_{θ_t} with improved performance in the domains seen by the robot during deployment. This process is constrained by three main challenges: (i) ground-truth labels of the target data are unavailable, (ii) the model must evolve online using the stream of images in $\mathcal{T}$, and (iii) the adaptation must be training-free, meaning that fine-tuning using backpropagation is not allowed.

Fig. 2 presents a general overview of the approach we propose for enabling training-free continual test-time adaptation in mobile robots. The framework builds on the training-free approach of SemLA [8], summarized in Section III-B. The core of our contribution, detailed in Section IV, is a domain shift detection method based on CLIP features to identify when adaptation is needed. By preventing unnecessary adaptation, our method reduces both the number of adaptations and the number of unique adapters used, while preserving segmentation performance, making the approach more efficient and practical for long-term, real-world deployments.

B. Training-free adaptation via adapters merging

Our approach for domain shift detection has been designed to adapt SemLA [8], a pipeline for training-free adaptation for VFMs, for online deployments in mobile robots. More specifically, SemLA consists of encoding domain-specific knowledge in adapters and dynamically merging the most relevant for the target domain in which the robot is operating, based on their proximity in CLIP's embedding space.

Let us consider a pre-trained model f_{θ_p} and a set $\mathcal{S} = \{\mathbf{S}_s \mid \mathbf{S}_s = \{\mathbf{I}_s, \mathbf{Y}_s, \mathbf{O}_s\}\}_{s=1}^{S}$ containing S *source* datasets. SemLA is a library of adapters formally described as

$$\mathcal{L} = \{(C_s, \theta_s) \mid s = 1, \dots, S\},$$

where C_s is the centroid descriptor for the dataset $\mathbf{S}_s$ (i.e., a vector computed by averaging the CLIP features extracted from the images $\mathbf{I}_s$), while θ_s denotes the weights of a low-rank adapter [19] attached to f_{θ_p} and trained on $\mathbf{S}_s$.

At deployment time, the library $\mathcal{L}$ provides training-free adaptation by selecting and merging the most relevant adapters for each frame, which are then attached to the model before inference. For each environment $\mathbf{E}_t \in \mathcal{T}$ and each image $I_i \in \mathbf{I}_t$, SemLA first computes the CLIP's feature embedding F_i for the test image. Then, this embedding is

used to retrieve the top-K most relevant adapters from the library $\mathcal{L}$, namely those adapters whose centroids are closest to the image's features in terms of Euclidean distance, calculated as $\|C_s - F_i\|_2$. These proximity scores of the selected adapters are then normalized using a softmax function and used to adjust their contribution to the image I_i. This is simply obtained by scaling the adapters' parameters using their relative normalized weights, formally $\theta_k^{'} = w_k \theta_k$, for $k = 1, \ldots, K$. Finally, the weighted adapters are merged and attached to the pre-trained model, which is then used for inference on I_i.

IV. Online Domain shift detection

One of the main limitations of applying SemLA on mobile robots is its per-frame adaptation. This requirement significantly compromises computational efficiency, as the model needs to be dynamically updated every iteration, combining multiple adapters. Furthermore, selecting new adapters at each frame can rapidly saturate GPU memory, as multiple adapters, often trained in similar domains, may be loaded for inference at a given frame but rarely reused in the subsequent ones.

Since robotic perception is highly correlated and domains change gradually over time, we propose a novel method for detecting domain shifts in the stream of images perceived by a robot. To reach this goal, we leverage the CLIP features already computed by SemLA for adapter indexing, and we monitor their variations over time to identify domain shifts and trigger adaptation only when needed, avoiding per-frame adaptation and improving efficiency.

Consider the ordered stream of images contained in all the target environments of $\mathcal{T}$, defined as $\mathbf{I}_\mathcal{T}$, with length $|\mathbf{I}_\mathcal{T}| = \sum_{t=1}^{T} |I_t|$ equal to the sum of the number of images of all environments $\mathbf{E}_t \in \mathcal{T}$. At each step i, we at first use CLIP to compute a feature embedding F_i for the image I_i. Then we compute the similarity of F_i with a feature memory $\mathcal{F}_{i-1}$, which is responsible for encoding, in CLIP embeddings' space, the visual appearance of the image stream until the last step $i-1$. $\mathcal{F}_{i-1}$ is updated using exponential moving average

$$\mathcal{F}_{i-1} = \alpha \mathcal{F}_{i-2} + (1-\alpha) F_{i-1}$$

until the last step $i-1$. The proximity score P_i between the embedding memory $\mathcal{F}_{i-1}$ and the current CLIP feature F_i is calculated as

$$P_i = 1 - \frac{\mathcal{F}_{i-1} \cdot F_i}{\|\mathcal{F}_{i-1}\|_2 \|F_i\|_2},$$

where $(\cdot)$ is the dot product. Practically speaking, P_i reflects how well the memory representation aligns with the current observation: lower values mean high similarity, while higher values indicate a change in the visual appearance.

Our method focuses on tracking the temporal evolution of P_i: an increasing trend suggests a domain change as the memory is becoming outdated, while a decreasing trend indicates a uniform and stable domain because incoming observations are more aligned with the memory. Let's call $P_{1\ldots i}$ the proximity indicators from the first to the current frame i. As shown in Fig. 2, this signal is noisy; therefore, monitoring domain shift from it can lead to spurious detections and unnecessary adaptations. To deal with this, we apply a low-pass filter obtaining a smoothed signal

$$\tilde{P}_{1\ldots i} = G_{(\sigma,r)} * P_{1\ldots i},$$

where $(*)$ is the convolution operation and $G_{(\sigma,r)}$ is a Gaussian kernel with standard deviation σ and radius r. At the current frame i, we compute the derivative of $\tilde{P}_{r\ldots i}$, so considering only the last $i-r+1$ values. It is calculated as

$$D_i = S_{(r)} * \tilde{P}_{r\ldots i},$$

where $S_{(r)}$ is a Sobel kernel with radius r. The trend of the signal $\tilde{P}_{1\ldots i-1}$ is then encoded in a memory variable $\mathcal{D}_{i-1}$, which accumulates derivatives until the last step $i-1$ using an exponential moving average. We compare the current derivative D_i with the memory $\mathcal{D}_{i-1}$. When the domain is changing, two conditions must be satisfied. First, the memory must be positive ($\mathcal{D}_{i-1} > 0$), indicating an increasing trend. Secondly, the current derivative must substantially exceed the memory, i.e., $D_i > \rho \mathcal{D}_{i-1}$, where ρ is a sensitivity hyperparameter. Instead of triggering the adaptation immediately, we wait until the derivative memory becomes negative ($\mathcal{D}_{i-1} < 0$), indicating that the signal trend is decreasing. This means waiting until the domain has settled, allowing the system to accumulate enough information for robustly representing the current visual domain. Following this, the descriptor used for adapter indexing is defined as the average of the CLIP features over the last $i-r+1$ frames, formally

$$F_{r\ldots i} = \frac{1}{i-r+1} \sum_{j=r}^{i} F_j,$$

that, compared to the most recent feature F_i, better captures the visual aspect of the current domain.

V. Experimental Evaluation

A. Experimental setup

To evaluate our TF-CTTA approach in open-vocabulary semantic segmentation under domain shift, we design a benchmark that combines a variety of indoor and outdoor datasets. The benchmark covers diverse scene types, label sets, and environmental changes (e.g., weather conditions) in order to better reflect long-term robotic perception under evolving domain conditions.

Datasets. Our benchmark consists of 2 indoor and 2 outdoor datasets.

We rely on Scannet [20] and Scannet++ [21], two large-scale indoor scene understanding datasets with diverse sensing modalities in multiple environment types, such as `apartment`, `office`, `classroom`, `bedroom`, etc. For these datasets, each environment is treated as a separate domain, resulting in 707 and 906 unique domains for Scannet and Scannet++, respectively. For Scannet, we use the widely adopted NYU40 label space [22], while for Scannet++, we consider the 100 most frequent object classes.

TABLE I

PERFORMANCE COMPARISON IN TRAINING-FREE CONTINUAL TEST-TIME ADAPTATION SETTINGS

		t →																
Test condition		Scannet → ACDC				Scannet++ → ACDC				Scannet → MUSES				Scannet++ → MUSES				**All Mean**
		$\mathbf{E}_1^{\mathbf{I}}$	$\mathbf{E}_2^{\mathbf{I}}$	$\mathbf{E}_3^{\mathbf{I}}$	$\mathbf{E}_4^{\mathbf{O}}$	$\mathbf{E}_5^{\mathbf{I}}$	$\mathbf{E}_6^{\mathbf{I}}$	$\mathbf{E}_7^{\mathbf{I}}$	$\mathbf{E}_8^{\mathbf{O}}$	$\mathbf{E}_9^{\mathbf{I}}$	$\mathbf{E}_{10}^{\mathbf{I}}$	$\mathbf{E}_{11}^{\mathbf{I}}$	$\mathbf{E}_{12}^{\mathbf{O}}$	$\mathbf{E}_{13}^{\mathbf{I}}$	$\mathbf{E}_{14}^{\mathbf{I}}$	$\mathbf{E}_{15}^{\mathbf{O}}$	$\mathbf{E}_{16}^{\mathbf{O}}$	
Zero-shot [3]	**mIoU**	39.0	36.8	36.3	39.7	35.7	33.6	32.3	32.6	32.1	31.8	31.6	31.3	30.7	30.1	29.7	29.6	33.3
	$\mathbf{N_A}$	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	$\mathbf{N_U}$	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
One-shot	**mIoU**	44.7	40.4	38.0	40.0	35.2	32.4	30.9	31.1	30.8	30.5	30.4	30.1	29.4	28.7	28.2	28.1	33.1
	$\mathbf{N_A}$	1	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
	$\mathbf{N_U}$	5	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Per-domain	**mIoU**	44.7	43.7	42.0	46.3	41.7	38.6	37.1	37.3	36.8	36.7	36.6	36.5	35.8	35.0	34.5	34.5	38.6
	$\mathbf{N_A}$	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	$\mathbf{N_U}$	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
Periodic	**mIoU**	**45.4**	43.7	**42.4**	46.4	41.8	39.1	37.6	37.8	37.3	37.2	37.0	36.8	36.2	35.4	34.9	34.8	39.0
	$\mathbf{N_A}$	5	5	6	9	13	13	16	8	5	5	5	3	14	15	14	3	9
	$\mathbf{N_U}$	16	16	17	8	39	38	47	8	16	14	16	6	42	44	43	7	24
Our	**mIoU**	45.3	**43.9**	42.2	**46.5**	**42.3**	**39.5**	**38.0**	**38.2**	**37.7**	**37.6**	**37.4**	**37.2**	**36.5**	**35.8**	**35.3**	**35.3**	**39.3**
	$\mathbf{N_A}$	6	6	7	10	16	18	22	9	5	5	6	2	18	20	19	2	11
	$\mathbf{N_U}$	15	14	16	7	35	37	44	7	14	13	16	6	39	42	38	6	22
SemLA [8]	**mIoU**	46.1	44.8	43.7	47.5	43.0	40.2	38.7	38.9	38.5	38.3	38.2	38.0	37.5	36.7	36.2	36.1	40.2
	$\mathbf{N_A}$	294	325	350	520	761	773	964	488	302	279	312	185	840	886	807	202	518
	$\mathbf{N_U}$	73	74	87	10	209	211	242	10	79	69	82	9	220	237	215	10	115

Results are averaged over 50 sequences of environments, where $\mathbf{E}_t^{\mathbf{I}}$ and $\mathbf{E}_t^{\mathbf{O}}$ refer to indoor and outdoor scenes randomly chosen in the considered datasets. The **best** and second best mIoU performance are evaluated without considering SemLA, which is the upper bound.

For outdoor contexts, we use ACDC [23] and MUSES [24], two popular driving datasets where domains are represented by different weather and illumination conditions. More precisely, ACDC has 5 different domains: `sun`, `night`, `rain`, `fog`, and `snow`. Differently, MUSES incorporates 8 domains in total, 4 weather conditions (`clear`, `rain`, `fog`, `snow`) in 2 different light conditions (`day`, `night`). Both datasets are evaluated using their 19 standard object categories.

Implementation details. We use SED [3] as the base model for open-vocabulary semantic segmentation, and we set $\sigma = 6$ and $r = 21$. As performed in [8], we attach a LoRA [19] to each linear layer of the visual encoder and the encoder of the main model, and then we train an adapter for each domain of the aforementioned datasets, which results in 1626 adapters, which are around 10 MB each. For ACDC and MUSES, we use the standard train/validation splits for each domain. For each environment of Scannet++, we use the iPhone and DSLR data acquired for different trajectories as the training and validation splits, respectively. In Scannet, since labels are provided at a high frequency (30 Hz), we compose the training split for each scene by sub-sampling an image every 10 frames, and for validation, we do the same with an initial offset of 5 frames.

Evaluation protocol and metrics. To validate our approach, we propose a benchmark in which the model is tested through a sequence of domains, reflecting our continual test-time scenario described in Section III-A.

1) We define a sequence $\mathcal{T}$ alternating 16 domains $\mathbf{E}_t$ randomly chosen from the 4 indoor/outdoor datasets previously mentioned.
2) We compose a new label space containing all the object categories needed by the 4 datasets (without repetitions), which is used for both model initialization and performance evaluation. By using a unified label space, we create a more realistic and challenging setting which better reflects real-world mobile robots' operation.
3) Evaluation is performed on the data used for adapter training in each $\mathbf{E}_t \in \mathcal{T}$. This is done as the validation split is a small amount of data to ensure convergence during training, so not compliant with a stream of robotic perceptions. As in [8], adapter selection is performed in a *leave-one-out* fashion: when adapting in $\mathbf{E}_t$, the corresponding adapter trained in $\mathbf{E}_t$ is not used, so the model is always tested on unseen data.
4) We evaluate cumulative performance across the entire sequence. To ensure fairness, the evaluation is repeated over 50 randomly sampled sequences.

Segmentation accuracy is computed using mIoU, while efficiency is measured considering, for each domain, how often the adaptation is triggered ($\mathrm{N_A}$) and the number of unique adapters used ($\mathrm{N_U}$). Minimizing these two indicators is associated with improving inference speed and reducing the GPU's memory usage, respectively.

Comparative methods. To evaluate our approach for domain shift detection in TF-CTTA, we compare with the following modalities for triggering adaptation:

- Zero-shot: the base model SED [3] without adaptation.
- One-shot: adaptation is triggered only in the first frame of the sequence $\mathcal{T}$.
- Per-domain: adaptation is triggered once per domain $\mathbf{E}_t \in \mathcal{T}$, at the first frame. For this baseline, we assume an oracle exists that correctly identifies where environments are concatenated.
- Periodic: starting from the first frame of the sequence, adaptation is triggered after a fixed interval of frame Δ_f, in our case we set $\Delta_f = 60$.
- SemLA [8]: the adaptation process is triggered at each frame, with no mechanisms for domain shift detection.

TABLE II

RESULTS EVALUATING SINGLE ENVIRONMENTS

Domains	ACDC [23] Means over 5 domains			MUSES [24] Means over 8 domains			Scannet [20] Means over 707 domains			Scannet++ [21] Means over 906 domains			**All means**		
	mIoU ↑	$\mathbf{N_A}$ ↓	$\mathbf{N_U}$ ↓	**mIoU** ↑	$\mathbf{N_A}$ ↓	$\mathbf{N_U}$ ↓	**mIoU** ↑	$\mathbf{N_A}$ ↓	$\mathbf{N_U}$ ↓	**mIoU** ↑	$\mathbf{N_A}$ ↓	$\mathbf{N_U}$ ↓	**mIoU** ↑	$\mathbf{N_A}$ ↓	$\mathbf{N_U}$ ↓
Zero-shot [3]	42.1	–	–	35.2	–	–	44.6	–	–	40.0	–	–	40.5	–	–
Per-domain	58.7	1	5	50.7	1	5	51.6	1	5	42.6	1	5	50.9	1	5
Periodic	<u>59.0</u>	8	8	<u>50.5</u>	4	6	<u>51.9</u>	6	18	<u>43.7</u>	14	44	<u>51.3</u>	8	19
Our method	**59.3**	10	8	**51.1**	4	7	**52.1**	7	17	**43.9**	19	41	**51.6**	10	18
SemLA [8]	59.6	480	10	51.3	188	10	53.2	326	80	44.6	839	228	52.2	458	82

Each domain is tested separately. Each dataset is evaluated using only its specific label space. Since in these settings One-shot and Per-domain represent the same baseline, One-Shot is omitted. The **best** and <u>second</u> best mIoU performance are evaluated without considering SemLA, which is the upper bound.

B. Results on training-free continual adaptation

Table I reports the results of our method and the baselines tested in a continual test-time manner. Overall, our method achieves segmentation accuracy (mIoU) close to the SemLA approach that, by performing a per-frame adaptation, represents a performance plateau. Despite this, our approach is remarkably more efficient in terms of the number of triggered adaptations (N_A) and the amount of used adapters (N_U).

On average, the naive One-shot approach achieves the lowest segmentation accuracy, even degrading performance compared to the Zero-shot baseline. Since adaptation is applied only to the first frame, performance improves only in the first environment and then rapidly degrades, suggesting that adaptation should be continuously performed along the sequence of images perceived by the robot during deployment. This is corroborated by the Per-domain baseline, where adapting the model at the beginning of each domain significantly improves performance. Despite promising, this setup is not realistic as it relies on an oracle capable of detecting the transitions between environments. The improvements provided by the Periodic baseline demonstrate that restricting adaptation only at the beginning of each environment is insufficient, as the visual appearance may vary even within a single scene. Compared to all these baselines, our method achieves the best performance in terms of segmentation accuracy, except for $\mathbf{E_1^I}$ and $\mathbf{E_3^I}$, in which Periodic leads to the best performance. This demonstrates that our approach is capable of identifying domain shifts in a stream of robotic sensor data for effectively triggering adaptation. Notably, regarding efficiency, our approach is comparable to Periodic. Despite triggering only 2 additional adaptations on average, our approach requires fewer adapters, leading to a lower memory footprint.

Although SemLA represents an upper bound in terms of segmentation accuracy, our method achieves comparable performance while remarkably improving efficiency. With only a marginal drop of 0.9 mIoU points compared to SemLA, our method triggers adaptation only 2% of the time and requires less than 20% of the adapters. From a computational perspective, considering a NVIDIA GeForce RTX 4080, the inference speed for SemLA (per-frame adaptation) is limited to approximately 2 FPS due to the overhead of loading adapters into the GPU, performing the merging procedure, and attaching them to the computational model graph. In practice, this throughput may further degrade on resource-constrained platforms such as NVIDIA Jetson family of devices, commonly used in robotic applications. In contrast, without adaptation, SED can perform inference at approximately 8 FPS, demonstrating that minimizing adaptation steps is crucial to ensure real-time operation.

C. Results on single environments

Since considering all the possible combinations of domains in continual settings is unfeasible, we conduct an additional experiment in which evaluation is performed separately for each environment.

The results, reported in Table II, corroborate the findings for the continual settings discussed in the previous section. While the Zero-shot baseline achieves the lowest segmentation accuracy, the Periodic strategy outperforms the Per-domain approach, indicating that adaptation within each domain still remains important. This is due to the presence of domain shifts within the same environment, such as transitions between rooms in indoor scenes or variations in visual conditions in outdoor scenes. Nevertheless, our method outperforms all baselines in terms of mIoU, demonstrating its ability to effectively identify domain shifts and trigger adaptation only when needed, while maintaining efficiency (in terms of N_A and N_U). Compared to SemLA, the performance gap is reduced from 0.9 to only 0.6 mIoU, while our method remains significantly more efficient. Finally, we remark that mIoU achieves higher values in this evaluation (Table II) than in continual settings (Table I) since each dataset is evaluated within its own label space. In contrast, the main evaluation in continual settings uses a more challenging unified label space across datasets, which better reflects the real-world operational conditions of robots.

VI. CONCLUSIONS

In this paper, we propose a training-free adaptation framework for continual test-time robotic scenarios. We introduce a domain shift detector leveraging CLIP features to accordingly trigger adaptation when needed, improving efficiency while maintaining task performance.

In future works, we will extend and improve our approach by considering also the semantic information encoded in CLIP features. Then, we plan to deploy the pipeline for training-free adaptation to validate its applicability in long-term robotic operation. Lastly, we want to extend the SemLA approach by proposing new strategies for training, indexing, and merging adapters with the purpose of improving segmentation precision, along with efficiency.

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